

Practical - 1 & 2

2520030030

U.Rohit Sai

Programm - 1

Code:

```
rohitsai@MSI: ~/OS
GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
int main()
{
    char cmd[100];
    printf("Enter Linux command: ");
    scanf("%s", cmd);
    pid_t pid = fork();
    if(pid == 0)
    {
        printf("Child PID : %d\n", getpid());
        execlp(cmd, cmd, NULL);
        perror("Execution Failed");
    }
    else if(pid > 0)
    {
        printf("Parent PID : %d\n", getpid());
        wait(NULL);
        printf("Child process completed.\n");
    }
    else
    {
        printf("Fork Failed\n");
    }

    return 0;
}
```

Output:

```
rohitsai@MSI: ~/OS
rohitsai@MSI:~$ cd OS
rohitsai@MSI:~/OS$ nano Linux.c
rohitsai@MSI:~/OS$ gcc Linux.c -o Linux
rohitsai@MSI:~/OS$ ./Linux
Enter Linux command: uname
Parent PID : 580
Child PID : 581
Linux
Child process completed.
```

Programm - 2

Code:

```
GNU nano 8.7.1
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
int main()
{
    int src, dest;
    char buffer[1024];
    int bytes;
    src = open("input.txt", O_RDONLY);
    if(src < 0)
    {
        printf("Cannot open source file\n");
        return 1;
    }
    dest = open("output.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);
    while((bytes = read(src, buffer, sizeof(buffer))) > 0)
    {
        write(dest, buffer, bytes);
    }
    close(src);
    close(dest);
    printf("File copied successfully.\n");
    return 0;
}
```

Output:

```
rohitsai@MSI:~/OS$ nano Linux.c
rohitsai@MSI:~/OS$ nano Copy.c
rohitsai@MSI:~/OS$ gcc Copy.c -o Copy
rohitsai@MSI:~/OS$ ./Copy
File copied successfully.
```